

1. Statement of Goals, Research Questions, and Specific Aims

The International Agency for Research on Cancer (IARC) classifies occupational exposure as a firefighter as a Group 1 known human carcinogen based on an abundance of evidence supporting a causal association between working as a firefighter and the development of cancer¹. Epidemiological studies have shown an increased incidence among firefighters of melanoma, mesothelioma, lung, gastrointestinal, kidney, bladder, prostate, brain, testicular, and hematologic cancers². The increased incidence of cancer among firefighters stems from their occupational exposures to a multitude of carcinogenic agents including products of combustion such as polycyclic aromatic hydrocarbons (PAHs) and volatile organic compounds (VOCs). These occupational exposures are in turn associated with multiple key characteristics of carcinogens including genotoxicity and epigenetic alterations^{1,3}. **Epigenetic alterations are included in the updated hallmarks of cancer.**⁴

Epigenetic investigations reveal inter-individual differences in biological responses and long-lived adaptations to carcinogenic chemicals influencing cancer risk. At the interface of the two-way interactions between genes and environment, DNA methylation (DNAm) is a covalent mark on cytosine-guanine (CpG) dinucleotides that can reflect cellular memory of repeated past exposures. Approximately 75% of blood DNAm differences between humans are estimated to be due to gene by environment interactions, aiding the molecular understanding inter-individual differences in biological responses.^{5,6} Genome-wide DNAm patterns can serve as intermediate biomarkers that link individual exposure measurements to disease outcomes and genetic differences.⁷ **DNAm signatures of exposures from products of combustion are under-investigated, particularly in firefighters.**

DNAm assays on cell free tumor DNA is an emerging exciting technology for its potential in the early detection by “liquid biopsy” of multiple cancer types, but it is understudied as a predictive biomarker for specific exposures in at-risk populations, including firefighters. We propose to develop a **firefighter early cancer screening (FECS) test using novel genomic and bioinformatic approaches.**

Among firefighters, live fire training instructors are faced with a particularly high burden of exposure to products of combustion because they facilitate dozens of training fires each year in addition to their normal firefighting duties⁸⁻¹¹. Co-investigator Thomas Sullivan (Los Angeles County firefighter) has participated in fire service community engaged research with the Los Angeles County Fire Department (LACoFD) as part of the Firefighter Cancer Cohort Study (FFCCS). Blood plasma samples were collected at the beginning and end of live fire training classes that last approximately one month for instructors exposed to 3-4 exposures per day, compared to recruits who had 1 exposure per day.

Our **goal** is to **reduce firefighters’ cancer risk by quantifying biological effects associated with exposure to products of combustion** and to develop protocols for **communicating research findings** to firefighters in a timely and comprehensible format that **motivates changes in their occupational health behaviors**. We will **leverage existing exposure data and biospecimens longitudinally collected from instructors and recruits during training fires** as well as from prior instructors to identify biomarkers of cancer risk from liquid biopsy. We will work with firefighters to **implement exposure reducing interventions during live fire training and test the effect of the intervention on our newly developed FECS liquid biopsy**. We **hypothesize** that exposure to carcinogenic agents from products of combustion will result in **quantifiable and dosage-sensitive changes** to circulating plasma DNAm of gene loci **relevant to risk for multiple cancer types**.

Aim 1. Identify DNAm regions sensitive to combustion products’ exposure and dose, prioritize DNAm regions for inclusion in FECS, and test the efficacy of each DNAm locus in FECS for intervention-sensitive changes. **1A.** DNA isolated from blood collected before and after exposures will be analyzed for DNAm changes genome-wide in Dr. LaSalle’s laboratory. **1B.** DNAm regions will be prioritized based on machine-learning prediction, overlap with cancer DNAm database, stability of changes in prior instructors, correlation with PAH and VOC levels, and replication using a different cohort and method. **1C.** FECS panel of ~10 prioritized DNAm loci will be tested for sensitivity in a new intervention cohort of instructors and recruits.

Aim 2. Develop culturally competent protocols for report back of research findings to firefighters. The impact on firefighters' perceived susceptibility/severity of occupational cancer risk will be gauged by surveys based on the Health Belief Model.

Aim 3. Design and implement exposure reducing interventions that are compatible with firefighters' public safety duties and result in detectable differences in FECS biomarker panel. Interventions will be disseminated to the fire service at large through California's FIRESCOPE Cancer Prevention Subcommittee and acceptability/feasibility will be gauged by surveys.

2. Background and significance

Occupational exposures of firefighters increase risk for a wide range of different cancers

California firefighters face significant occupational risks for developing various types of cancer, due to prolonged exposure to carcinogenic substances encountered during urban and wildland firefighting activities, as well as simulated exposures during instruction and training. Firefighters are at a notably higher risk for multiple cancers compared to the general population. California firefighters (1988-2007) showed elevated risk for multiple myeloma (Odds ratio OR 1.4), acute myeloid leukemia (OR 1.4), cancers of the esophagus (OR 1.6), prostate (OR 1.5), brain (OR 1.5), and kidney (OR 1.3).² High rates for multiple myeloma (OR 1.53) and non-Hodgkin's lymphoma (OR 1.51) were observed in a meta-analysis of 32 firefighter studies across multiple regions.¹² A US based study (1950-2009) examined exposure-response relationships between surrogates of firefighting exposure and found that lung cancer and leukemia mortality risks were modestly increasing with firefighter exposures.¹³ These findings add to evidence of a causal association between firefighting and cancer. These heightened risks are largely attributed to inhalation of toxic smoke, absorption of hazardous chemicals through the skin, and exposure to asbestos, benzene, and formaldehyde, which are prevalent in burning materials and firefighting foam. These findings highlight the critical importance of ongoing research and policy interventions aimed at mitigating cancer risks in this vulnerable occupational group.

Early cancer detection improves survival, but most existing screens are cancer-specific

Early detection of cancer is a crucial determinant in improving survival rates and reducing mortality. Early-stage cancers are often more treatable and responsive to less aggressive therapies, which significantly enhances patient outcomes.¹⁴ For instance, the five-year survival rate for localized breast cancer is approximately 99%, compared to only 27% when it has metastasized. Similarly, the survival rate for localized colorectal cancer is about 90%, but it drops to 14% when detected at an advanced stage. These statistics underscore the importance of routine screening programs, which can identify cancer in its nascent stages before symptoms manifest, thereby allowing for timely and potentially life-saving interventions. Effective screening techniques such as mammography, colonoscopy, and low-dose computed tomography have been instrumental in reducing mortality rates for breast, colorectal, and lung cancers, respectively.

However, current cancer screening programs face significant challenges, particularly for individuals with increased risk due to occupational exposures. Most screening protocols are designed to detect individual cancer types rather than providing a comprehensive assessment for multiple cancers, which is a limitation for high-risk populations such as firefighters who are exposed to a variety of carcinogens. Additionally, these cancer screening programs often begin screening at ages that might be too late for those who have been exposed to occupational hazards since early adulthood. For example, firefighters exposed to benzene, asbestos, VOCs, and PAHs have elevated risks for developing multiple cancer types. There is a critical need for novel screening programs tailored specifically to firefighters that would commence earlier and cover a broader spectrum of cancer types. Enhanced surveillance and personalized risk assessment tools could bridge this gap, ensuring that those with significant occupational exposures receive timely and comprehensive cancer screenings. Personalized risk assessment tools would also have the potential to inform firefighters' perceived susceptibility/severity of occupational cancer risk and increase the likelihood of administrative interventions that could reduce dosage of exposures during controllable events including firefighter training protocols.

Cell free DNA screening tests have moved from prenatal genetic screens in pregnancy to detecting tumor DNA in cancer patients

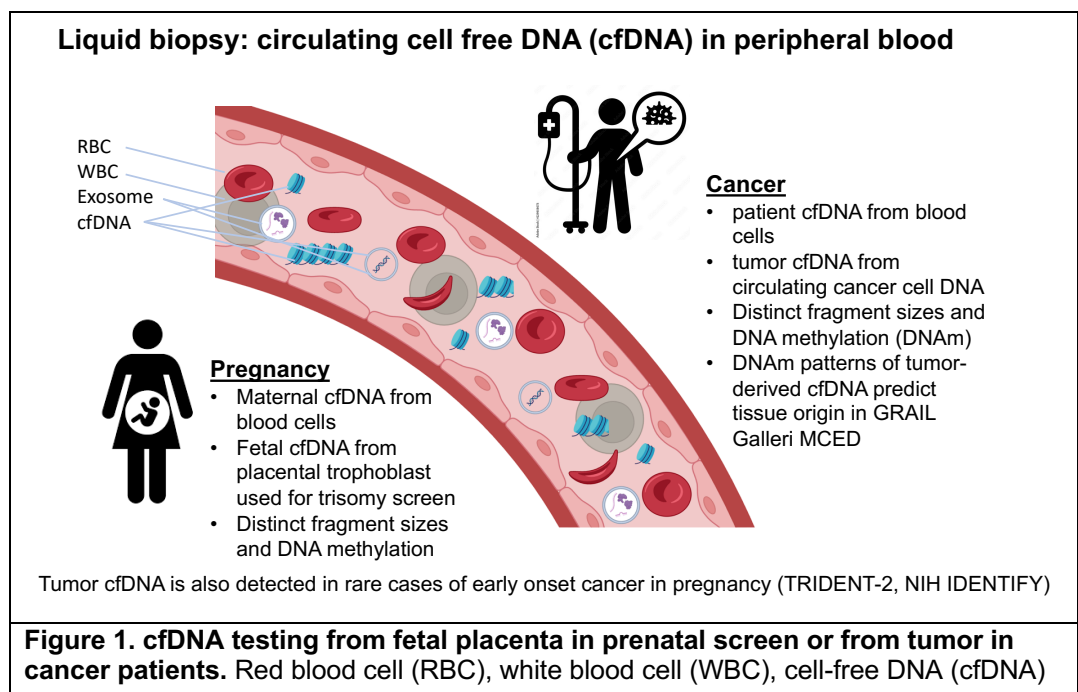
Cell-free DNA (cfDNA) screening tests originally emerged in the context of prenatal genetic screening during pregnancy, revolutionizing the way genetic disorders are detected in fetuses (**Figure 1**).^{34,35,36} These non-invasive prenatal tests (NIPTs) analyze cfDNA fragments that circulate in a pregnant woman's blood, which include DNA from the placenta that is representative of the fetal genome. Genetic evidence from cases of anembryonic pregnancies or confined placental mosaicism have demonstrated that cffDNA originates from the trophoblasts of the placenta.^{37–39} Introduced in 2011, these tests quickly gained popularity due to their high sensitivity and specificity for detecting chromosomal abnormalities such as trisomy 21 (Down syndrome), trisomy 18 (Edwards syndrome), and trisomy 13 (Patau syndrome).¹⁵ A recent unexpected development from screens of routine NIPT are that some early-stage cancers are being detected from routine NIPT screens in otherwise healthy pregnant women. In the Dutch TRIDENT 2 study, pregnant women with abnormal NIPT results followed by normal karyotype and ultrasound (0.03%) were followed by cancer screening (whole body MRI) and ~50% had a detectable chromosomal aberration and/or malignancy¹⁶.

Epigenetic evidence from genome-wide DNAm investigations has confirmed the placental origin of fetal cfDNA via detection of hypomethylated domains called partially methylated domains,^{40,41} which are uniquely characteristic of placenta in individuals without cancer.^{30,42} Interestingly, cancer DNA

methylome characteristics of global hypomethylation and promoter hypermethylation are highly similar to placenta, reflecting similar biological functions.^{17,18} Fetal trophoblast cells of the placenta invade the maternal uterine lining and spiral arteries, performing angiogenesis, similar to the invasive properties of cancer cells.¹⁹ Human tumors aberrantly express placental-specific genes, and together with germline-specific genes, placental genes can be biomarkers of aggressive metastasis in cancer.²⁰ Building on the success of cfDNA technology in prenatal screening, research studies have extended its application to oncology for the detection of circulating tumor DNA (ctDNA) in cancer patients. ctDNA refers to the fraction of cfDNA that originates from cancer cells, carrying tumor-specific genetic and epigenetic alterations. This advancement has enabled the development of liquid biopsy tests, which offer a non-invasive method to detect, monitor, and manage cancer. The transition from prenatal genetic screening to cancer detection underscores the versatility and transformative potential of cfDNA analysis in personalized medicine that are well poised to move from genetic to epigenetic testing.

Multi-cancer early detection tests from liquid biopsy samples utilize differences in DNA methylation

Multi-cancer early detection (MCED) tests utilizing liquid biopsy blood samples represent a transformative advancement in oncology, capitalizing on the unique patterns of DNA methylation associated with different cancer types as well as their tissue of origin.^{21,22} These tests analyze cfDNA



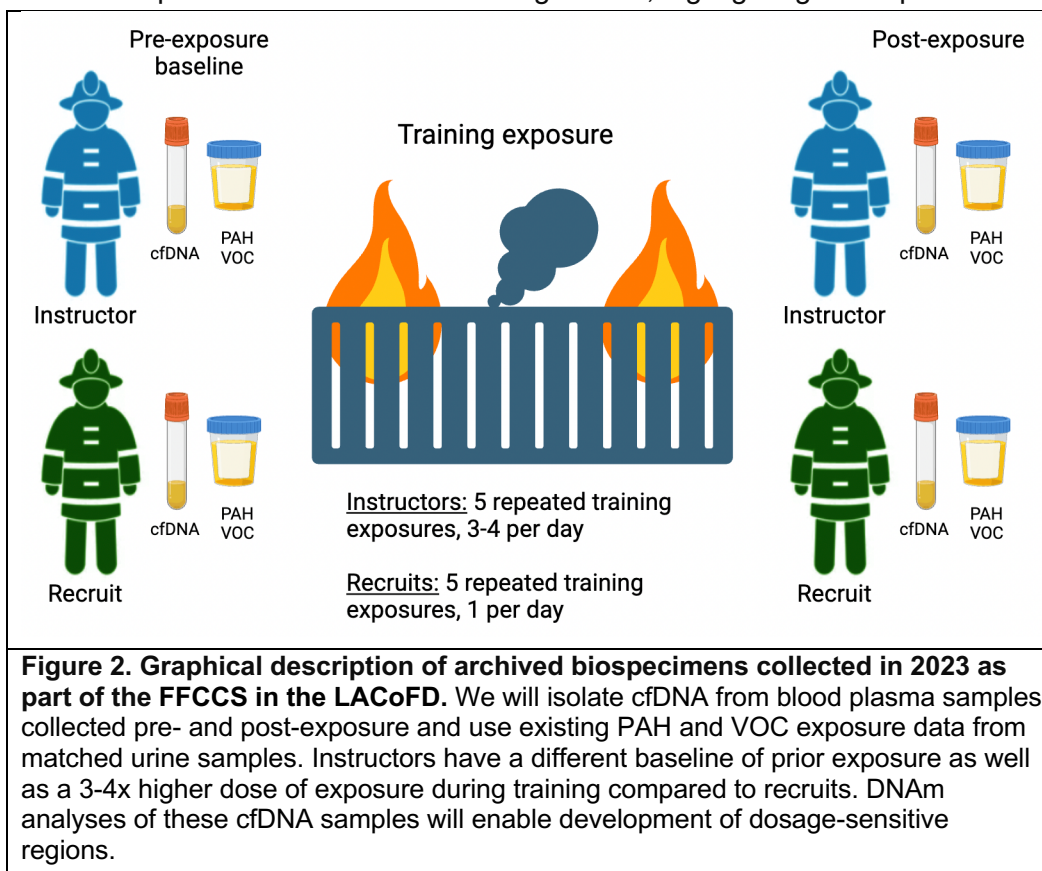
circulating in the bloodstream, identifying aberrant methylation patterns that are indicative of malignancy. A study comparing machine-learning predictors in the development stage of MCED tests demonstrated that whole genome methylation patterns were better predictors than genetic tests.²³ By detecting these cancer-specific methylation signatures, MCED tests can identify multiple types of cancer from a single blood sample, offering a non-invasive and comprehensive screening approach.

One prominent example of this technology is the Illumina Galleri test, developed by GRAIL.²⁴ The Galleri test analyzes methylation patterns in cfDNA to detect over 50 types of cancer, often before symptoms appear. Clinical studies have demonstrated its high specificity and sensitivity, particularly for cancers that currently lack effective screening methods, such as pancreatic cancers. Other similar tests in clinical trials include those from companies like Freenome and Guardant Health, which also leverage advanced methylation profiling techniques to identify cancer-specific DNA alterations.²⁵ These MCED tests have shown promise in early detection and intervention, potentially reducing cancer mortality by enabling earlier and more targeted treatment strategies. As these technologies advance and undergo further validation in clinical settings, they hold the potential to revolutionize cancer screening and management. However, MCED tests being developed are trained on patients who already have cancer, not specific at-risk populations who may develop cancer in the future because of the occupational exposures, including firefighters.

A firefighter early cancer screening (FECS) blood test can be developed from DNAm patterns

Environmental exposures can leave long-lasting alterations in DNAm patterns throughout the genome that can be potentially used to assess future cancer risk. DNA methylation changes in human tissues can serve as one of the earliest markers of carcinogenesis, appearing not only in the tumors themselves but also in adjacent normal-appearing tissues.²⁶ This phenomenon, known as the "field effect" or "field cancerization," reflects the widespread epigenetic alterations that occur during the early stages of cancer development. For examples, methylation of the CpG island in the promoter region of the p16 tumor suppressor gene was present in non-tumor colonic mucosa of patients with colorectal cancer, suggesting a field defect in these tissues.²⁷ Similarly, methylation of the p16 and MGMT genes occurred in the normal bronchial epithelium of smokers with lung cancer, highlighting the impact of

environmental carcinogens on widespread epigenetic alterations.²⁸ These studies underscore the potential of DNA methylation changes in tissues damaged by exposures that are likely to be released into the bloodstream as cfDNA and therefore serve as early biomarkers for cancer detection and risk assessment. There is a distinct need for comprehensive screening strategies that consider these early molecular



alterations of the earliest cancer stages in firefighters that have enhanced cancer risks due to their repeated exposures.

Together, these prior studies solidify a strong premise that a novel MCEd test could be developed by genome-wide DNA methylation signatures specifically relevant to firefighter exposures. This proposal is taking advantage of previously collected and archived biospecimens in a “natural experiment” of firefighter training. **Figure 2** graphically demonstrates the overall study design for archived samples and data that will be used for the initial development of the firefighter early cancer screen (FECS).

This longitudinal experimental design will allow us to identify genomic regions that have quantifiable biological differences in DNAm following repeated exposures to products of combustion. Using this approach, we expect to identify thousands of DNAm differences between pre- and post-exposures at two different doses (Instructor-high, Recruit-low), as well as between Instructors and Recruits at both time points. These dosage-sensitive regions will be further selected for prioritization by machine-learning approaches, comparison to known cancer DNAm alterations, stability of DNAm changes in prior instructors, correlation with PAH and VOC levels, and replication in the Climate Action cohort DNAm data. From these analyses, we will develop a FECS panel of ~10 prioritized regions that we will test on plasma biospecimens collected in a new cohort of Instructors at LACoFD that will include an intervention of either one training exposure per day compared to the standard 3-4 exposures per day.

This overall plan, outlined in more detail below, will enable a community-directed intervention approach to be incorporated into the development of the FECS testing protocol in a well-defined cohort with measurable exposures to products of combustion in a natural experiment.

Experience, knowledge, and skills of the research team

The collaborative research teams at UC Davis and LACoFD have been working together for the past year on the UCOP Climate Action grant entitled “Exposure assessment, health monitoring, and cancer control in wildland firefighters” (UCD: Hussain, PI & LaSalle co-I; LACoFD: Sullivan, firefighter research assistant). The biospecimens described in **Figure 2** were collected at LACoFD. In addition, multiple data and biospecimen collections are already being performed at LACoFD for the UCD-LACoFD Climate Action grant. The Climate Action grant is complementary but not overlapping with the current proposal which uniquely focuses on developing a cfDNAm-based multi-cancer FECS.

Janine LaSalle, Ph.D., UC Davis Co-Principal Investigator: Dr. LaSalle is a Professor of Microbiology and Immunology and a member of the Comprehensive Cancer Center at UC Davis School of Medicine. Dr. LaSalle is an interdisciplinary scientist with expertise in epigenetics and gene x environment interactions, as well as whole genome DNAm sequencing and bioinformatic technologies. Dr. LaSalle is the Deputy Director of the UCD Environmental Health Sciences Center and a member of the congressionally-supported UCD Health Institute on Wildfires, Health, Equity, and Resilience (described below). She also serves as co-Director of the Perinatal Origins of Disparities Center, was a prior Associate Director of the Genome Center at UC Davis. Dr. LaSalle will lead the research in this project related to the development of the cfDNAm-based multi-cancer FECS, mentor Mr. Williams and Ms. Thrall, and participate in quarterly report-back seminars and discussions about interventions with LACoFD firefighter in collaboration with co-PI Mr. Sullivan.

Thomas Sullivan, LACoFD Co-Principal Investigator: Thomas R Sullivan is a Firefighter Paramedic in East Los Angeles and has been in the EMS/ Fire Service field since 2011. Firefighter Sullivan is a training officer and state rater for the Training Services Section, and has instructed EMT classes throughout LA County. For years Firefighter Sullivan has been instrumental in working with the Firefighter Cancer Chart Study in organizing multiple community based sample collection and follow up sample collection for all studies in LA County as it pertains to the cancer studies LACoFD has been involved in. Firefighter Sullivan has also been responsible for orchestrating introduction presentations for the goals of all sample collection and has assisted in dissemination of information and data of the studies back to fire service personnel.

Shehnaz Hussain, Ph.D., UC Davis Co-Investigator: Shehnaz K. Hussain is Professor, UC Davis School of Medicine, Director, Office of Population Health, and Associate Director of Population Sciences, UC Davis Comprehensive Cancer Center (UCDCCC). She is a molecular epidemiologist with

~15 years of leadership experience in the design, conduct, and analysis of cohort studies with a focus on modifiable risk factors and immunity. Dr. Hussain will provide epidemiological expertise and the sharing of data and samples from the cohort of firefighters from the UCOP Climate Action grant.

Logan Williams, UC Davis graduate student: Mr. Williams is a Ph.D. candidate in the lab of Dr. LaSalle, where he is pursuing his doctorate in Integrative Genetics and Genomics. Prior experiences studying cfDNA in pregnant women as well as optimizing liquid biopsy platforms at Roche Sequencing Solutions have allowed Mr. Williams to develop proficiency in the extraction and sequencing of cfDNA, even from suboptimal sample conditions. Mr. Williams will oversee and execute upon the “wet lab” portion of this project, isolating all cfDNA samples from participant plasma and preparing these samples for whole-genome bisulfite sequencing. Applying bioinformatic skills he has developed both in prior research and in Dr. LaSalle’s lab, Mr. Williams will also analyze all collected sequencing data, identifying regions of DNA responsive to combustion products and contributing to the selection of top candidate regions for FECS development.

Emily Thrall, UC Davis Junior Specialist: Ms. Thrall has worked as an undergraduate intern and scholar in the UC LEADS program over the past two years. She is on track to complete her B.S. in Biomedical Engineering from UC Davis in December 2024 and will be hired back in the LaSalle lab as a full time Junior Specialist after graduation. Ms. Thrall has obtained expertise in biospecimen labelling and handling including FreezerWorks and biohazard training. She also has gained expertise in DNA isolation, data management, and bioinformatic analyses including creating data visualization and analysis in R. Her role on the project will be to perform all wet bench and bioinformatic analyses on the project under the mentorship of Dr. LaSalle and Mr. Williams. In addition to her laboratory research, Ms. Thrall will travel ~4 times to LACoFD to assist Mr. Sullivan in biospecimen labelling and shipping, as well as community engagement activities. As the daughter of a LACoFD Captain and a scientist, she is well suited for answering questions from firefighters about the research project.

The UC Davis Environmental Health Sciences Center provides critical programs and services to help researchers studying environmental factors that influence health. High-quality scientific evidence and, especially, new knowledge, can be used to find ways to reduce harmful exposures or lessen their impact on people, which translates into improved public health. The Center also works with communities impacted by harmful exposures to chemicals or particles, bringing their priorities to scientists so that the research is most relevant to those most affected.

The UC Davis Health Institute on Wildfires, Health, Equity, and Resilience addresses the challenges of wildfires with a cohesive, comprehensive and impactful program. To date, the team has launched longitudinal studies, collected biologic specimens, and begun examining both respiratory and mental health outcomes in wildfire survivors. New congressional support in the 2024 US budget was obtained to enable the building of a unique combustion and exposure facility for testing various fuel types to determine the composition of the smoke, and then to perform experimental studies in animals that will have direct relevance to people. This congressional earmark, supported by the late Senator Diane Feinstein, also allows establishment of mobile wildfire health clinics that will reach underserved communities and those in rural and remote areas, while enhancing the knowledge gained, by including physical examinations. Laboratory and computational equipment are also included that will be instrumental in future research on the health impacts of products of combustion at UC Davis.

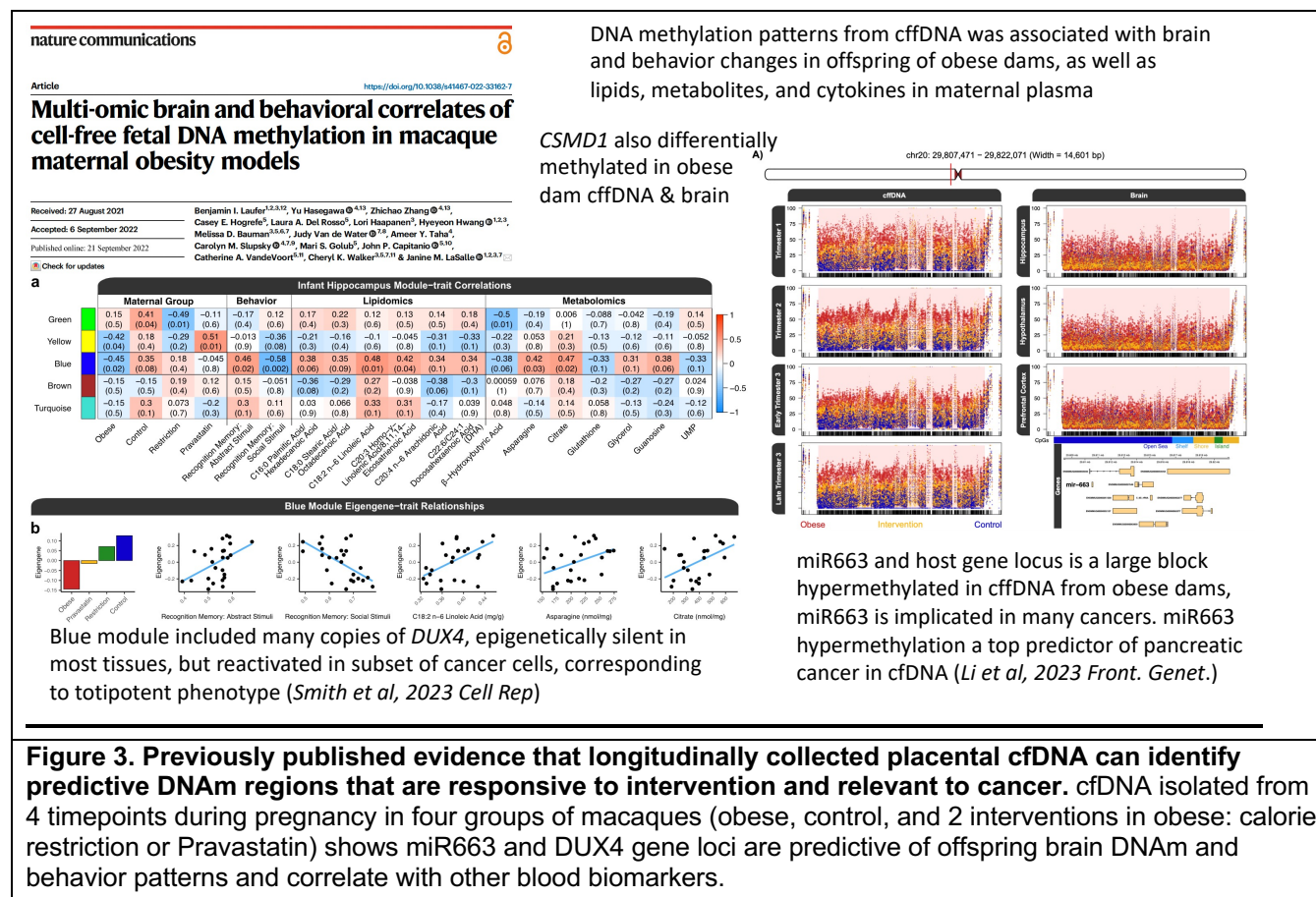
Preliminary Data

Preliminary data from the UCOP Climate Action Grant. The California Firefighter Clinical Research Site (CAFF CRS) successfully recruited 294 participants in Year 1 of the Climate Action grant. These included 32 new Recruits and 262 Incumbent firefighters. Recruits were younger (mean age 28.7 vs 39.2), slightly less male (93.8% vs 95.8%), more ethnically and racially diverse, and less likely to have obesity or have ever smoked cigarettes compared to Incumbents. Recruits had a mean of 0.8 years of firefighting compared to 13 years for the Incumbents. DNAm data from whole blood samples assayed by array-based methods will be generated within the next year and utilized for comparisons to gene loci identified by cfDNA in Aim 1B of the current proposal.

Preliminary data supporting the detection of cancer-relevant DNAm signatures in cell-free DNA

Dr. LaSalle's research team has been at the forefront of developing whole genome bisulfite sequencing (WGBS) and bioinformatic analysis tools to investigate DNAm signatures of human neurodevelopmental disorders such as autism spectrum disorder. An overlap between autism and cancer risk genes and pathways²⁹ was recognized by Dr. LaSalle and coauthors in 2016, and this convergence has continued to be recognized by genomic and system biology approaches.^{30,31} In 2013, Dr. LaSalle's laboratory published the first whole genome sequence of the placental methylome, showing that it was epigenetically most similar to cancer in the presence of partially methylated domains.³² The published and preliminary data shown below has been using cfDNAm analyses derived from placenta during pregnancy, but the relevance of these findings to cancer will also be presented.

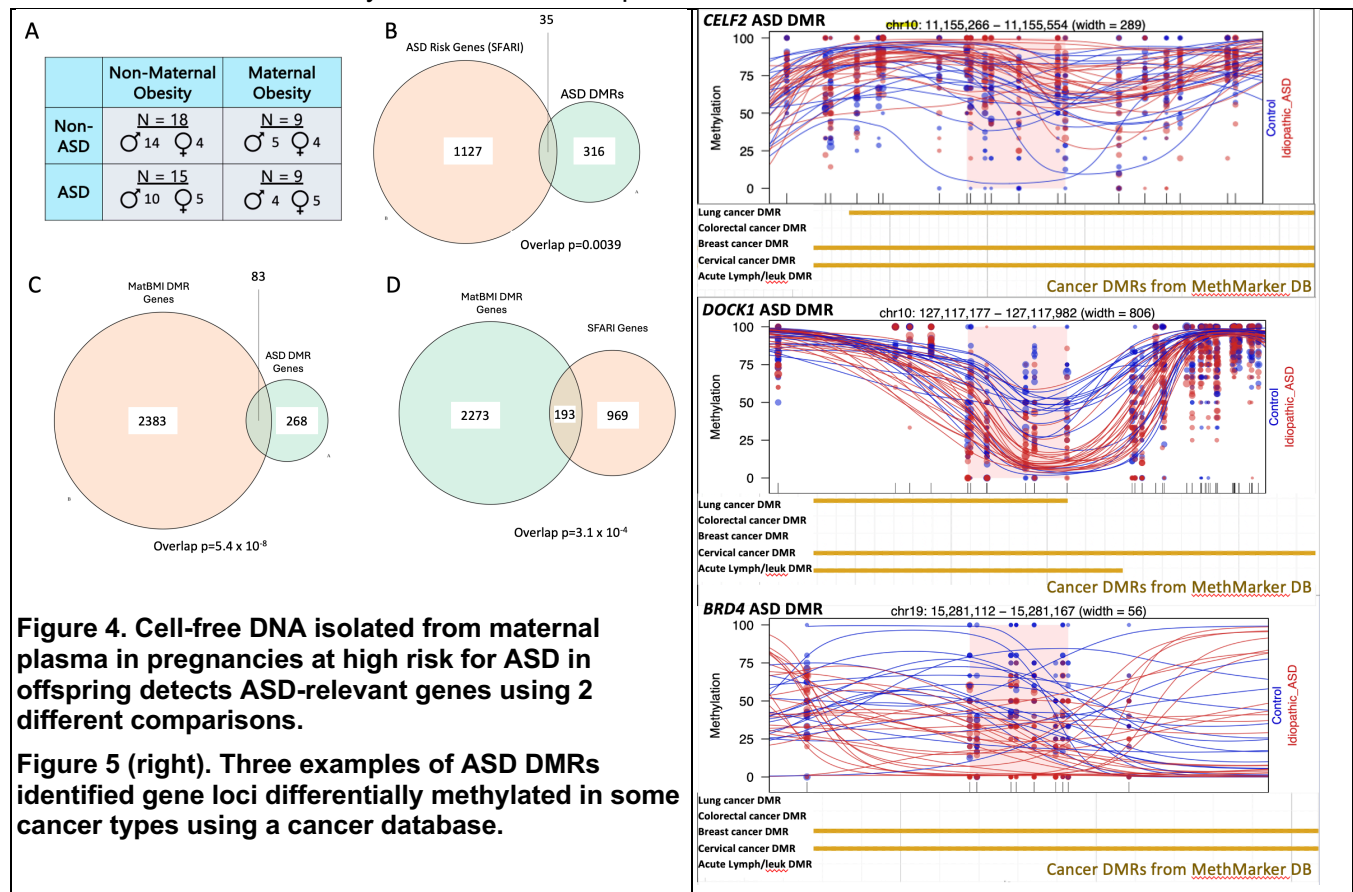
DNA methylation patterns in maternal cell free DNA are predictive of offspring outcomes. DNA methylation assays on cell-free tumor DNA is an emerging exciting technology for its potential in the early detection of multiple cancer types³³, but it is understudied in prenatal samples. Dr. LaSalle's lab was the first to demonstrate that DNA methylation patterns from cell free DNA could be predictive of offspring brain DNA methylation and social memory, following exposure to maternal obesity (**Figure 3**).³⁴ The two major epigenetically dysregulated genes and gene pathways identified from cffDNA of obese dams were a large differentially methylated block of multiple DMRs around the miR663 locus and the multicopy DUX4 co-methylated transcriptional pathway, both which also have been identified as epigenetic predictors of cancer.^{35,36} Together, these findings support the premise that perinatal exposures and metabolic conditions including obesity leave patterns of DNA methylation in cell free fetal DNA relevant to cancer risk in later life.



Preliminary data from cfDNA on archived human plasma samples from the ASD-risk MARBLES cohort

Dr. LaSalle's lab has followed up on the results from the macaque model of maternal obesity in unpublished results utilizing archive maternal plasma samples (0.5-1 ml) from the prospective human cohort Markers of Autism Risk in Babies, Learning Early Signs (MARBLES). Mr. Logan Williams

successfully isolated cfDNA from a total of 51 pregnancies and performed whole genome bisulfite sequencing (WGBS). **Figure 4A** shows the cross-comparison design for generating differentially methylated regions (DMRs) that is similar to the 4-way comparison approach shown in Figure 2 for the proposed study. When comparing ASD to control (Non-ASD), he identified 316 ASD DMRs mapped to genes that significantly overlapped with a curated set of known ASD risk genes (Fig 4B). A much larger number of DMRs (2,383) were identified when comparing samples from mothers with or without obesity (Fig. 4C), which is not surprising considering the cfDNA collected during pregnancy is mostly maternal in origin. Remarkably, there was still a highly significant overlap (by Fisher's exact) in the ASD DMR gene list compared to that of the Maternal BMI DMRs, as well as comparing genes in common to Maternal BMI DMRs and known ASD risk genes (SFARI)(Fig 4C-D). These results demonstrate that despite the daunting challenge of maternal cfDNA containing a mixture of different sources, we were still able to identify a clear ASD signature from a risk-associated exposure, in this case maternal obesity which increases ASD risk by ~36%.³⁷ These exposure-sensitive ASD DMRs are also relevant to



cancer, for which obesity is also a major risk factor. Examples are shown in **Figure 5** for three of the ASD DMR genes in common with maternal BMI DMRs and SFARI ASD risk genes. Each graph in Fig 5 plots % methylation of each CpG for each sample on the y axes (red, ASD; blue, non-ASD; lines show smoothed values). The same chromosomal regions were queried in the MethMarker database for known cancer DMRs³⁸ (yellow bars on bottom tracks). Remarkably, all 3 genes were also associated with DMRs in multiple specific cancer types, although the combination of cancer types was different for each gene. All 3 gene have been implicated in both ASD and cancer. As examples, *CEL2F2* is a possible prognostic indicator of both lung and breast cancers³⁹, increased *DOCK1* expression is a negative prognostic indicator of treatment in lymphoma⁴⁰, and *BRD4* is directly targeted by inhibitors in clinical trials for multiple cancers⁴¹. These results emphasize the myth-busting reality that genes are not disease specific, and that common exposures (such as obesity and smoking) interacting with common genetic variation occur over a convergent set of epigenetically sensitive genes involved in multiple disease types. Therefore, in this proposal we will be identifying epigenetic sensitive markers of exposure (products of combustion) that are expected to be relevant to predicting future cancers.

3. Research Methodology: Research Design, Conceptual Framework, and Data Analysis

Aim 1. Identify DNAm regions sensitive to combustion products' exposure and dose, prioritize DNAm regions for inclusion in FECS, and test the efficacy of each DNAm locus in FECS for intervention-sensitive changes. (LaSalle lab, UC Davis)

Hypothesis: Firefighter instructors' exposure to carcinogenic agents from products of combustion during training fires will result in quantifiable and dosage-sensitive changes to circulating plasma DNA methylation of gene loci relevant to risk for multiple cancer types and stably maintained after exposure.

Rationale: Unlike transcriptional responses to exposures, which are immediate and often transitory, DNA methylation (DNAm) marks are more stable, and therefore have the potential to be both reflective and predictive of longer-term health outcomes.⁴² Furthermore, DNAm is highly abundant, correlates with other epigenetic layers including open chromatin and histone modifications, and is the layer closest to and most influenced by DNA sequence.⁴³ DNA methylation analysis of cell free tumor DNA is an emerging exciting technology for its potential in the early detection of multiple cancer types³³, but it is understudied as a predictive biomarker for specific exposures in firefighters. **Figure 6** outlines the workflow of the three subaims that seek to identify dosage-sensitive DNAm signature of firefighters' exposures (1A), prioritize top gene loci for FECS inclusion (1B), and test a FECS panel of ~10 regions on the new instructor intervention cohort from Aim 3.

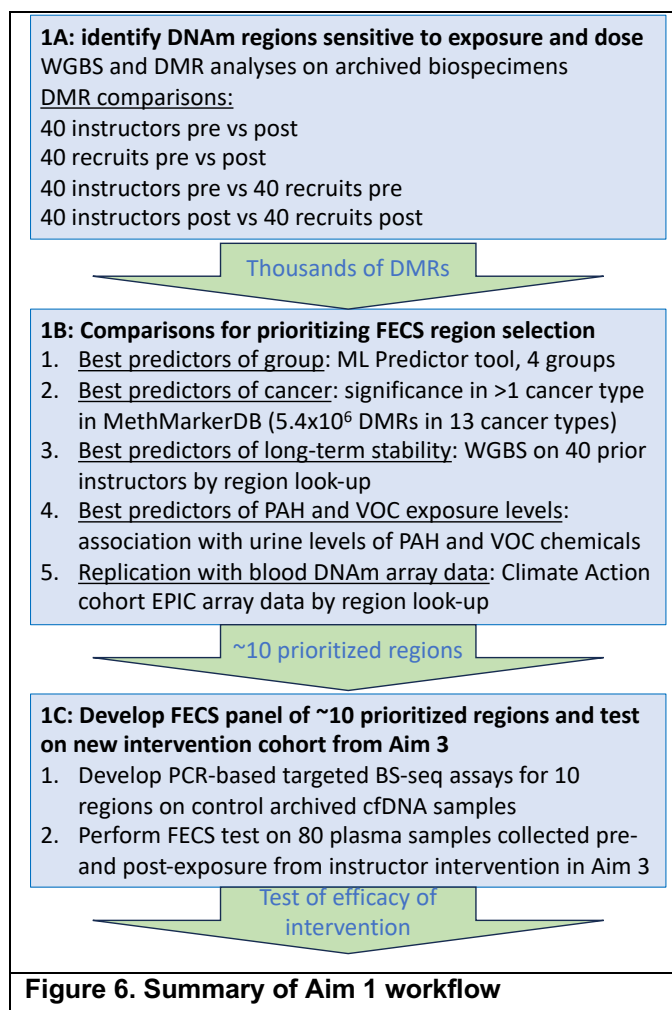
Aim 1A. DNA isolated from blood collected before and after exposures will be analyzed for DNAm changes genome-wide in Dr. LaSalle's laboratory.

Biospecimens: Figure 2 graphically demonstrates the archived samples and data that will be used for the initial development of the firefighter early cancer screen (FECS). Approximately 160 blood plasma samples that were collected pre- and post-exposure from 40 training instructors and 40 recruits in January 2023 and 40 samples from former instructors in the FFCCS will be sent to Dr. LaSalle's lab for cell free DNA isolation.

DNA isolation: As described previously³⁴, 10-15 ng of cell free DNA will be isolated from 1-2 ml of frozen human plasma using the QIAamp Circulating Nucleic Acids kit.

Whole genome bisulfite sequencing (WGBS): WGBS libraries will be constructed from ~10 ng DNA and bar coded using the xGen Methyl-Seq Library Prep Kit and xGen UDI Primers. Libraries will be sequenced on an Illumina Novaseq X flow cell to generate paired 150 bp reads, single indexed. This will generate ~400 million reads (12x coverage) per sample, sufficient to detect methylation of both large- and smaller scale regions. Sequences will be processed in the CpG_Me bioinformatic pipeline, developed by Dr. LaSalle's lab⁴⁴⁻⁴⁶, using Trim Galore, FastQ Screen⁴⁷ and Bismark⁴⁸ tools to perform quality control and alignment.

Differential methylated region (DMR) analyses using DMRichR: To find gene and chromosomal locus-specific DNAm changes between pre- and post-exposures in cell free DNA, small differentially methylated regions (DMR) and large blocks will be identified, as previously described.^{44,46} We will first identify DMRs comparing pre- and post-exposure cfDNA



for instructors and recruits separately for the main effect of exposure, adjusted for age and smoking status using the dmrseq R package⁴⁹ as part of DMRichR pipeline.^{44,45} We will then identify DMRs comparing instructors to recruits separately for pre- and post- cfDNA. Based on prior analyses of similar sizes, we expect to identify several hundred to possibly thousands of DMRs in each of the four comparison groups. These will be compared and prioritized in Aim 1B. All DMRs will be mapped to genes and examined for gene function and biological pathway enrichments using GREAT and KEGG pipelines.

Aim 1B. DNAm regions will be prioritized based on machine-learning prediction, overlap with cancer DNAm database, stability of changes in prior instructors, correlation with PAH and VOC levels, and replication using a different cohort and method.

Prioritizing best predictors of exposure and dosage using machine learning tools: DMRichR has built-in machine learning predictors using two different algorithms. Random Forest (RF) DMR importance ranking is from Boruta algorithm in *Boruta* package, while support vector machine (SVM) DMR importance ranking is from *sigFeature* package. An example from the ASD DMRs shown in Fig 4 list of common top DMRs is shown in **Figure 7**. For each of the 4 group comparisons, we will obtain the top 10 predictors by both algorithms as well as the common predictors between them. These lists will provide unbiased prioritization of the best predictors cfDNA belonging to each group (instructor, recruit, pre-, post). We expect there to be some overlap between instructors and recruits post-exposure.

Annotations of common DMRs									
common DMRs from top 10 DMRs of two DMR importance ranking lists from RF and SVM algorithms									
RF.rank	SVM.rank	seqnames	start	end	width	annotation	geneLength	SYMBOL	GENENAME
2	2	chr10	36811014	36811716	703	Distal Intergenic	106970	ANKRD30A	ankyrin repeat domain 30A
4	3	chr1	230155092	230155401	310	Intron (ENST00000494106.1/2590, intron 1 of 6)	185438	GALNT2	polypeptide N-acetylgalactosaminyltransferase
8	6	chr9	96688696	96688939	244	Intron (ENST00000661289.1/ENST00000661289.1, intron 1 of 2)	15741	PRXL2C	peroxiredoxin like 2C
10	10	chr8	107366477	107367082	606	Intron (ENST00000517746.6/284, intron 1 of 8)	248437	ANGPT1	angiotensinogen 1

Figure 7. Example of output from common DMRs from the top 10 DMRs of two importance ranking lists from Random Forest (RM) and Support Vector Machine (SVM) algorithms

Prioritizing best predictors of cancer relevance: MethMarker DB is a published and curated cancer database which includes 658 WGBS datasets, 724 curated DNA methylation biomarker genes from 1425 published articles, and 5.4 million DMRs from 13 common types of cancer as candidate DNA methylation biomarkers.³⁸ We will compare our DMR lists from each of the four comparison groups to MethMarker DB for overlap with cancer DNAm biomarkers and prioritize those with relevance to at least one cancer type with increased risk in firefighters. This list will be further prioritized for those DMRs that also match those prioritized by at least one of the machine learning tools.

Prioritizing best predictors of long-term stability: In the WGBS data obtained from the 40 prior instructors, we will perform a region look-up analysis for DNA methylation levels within prioritized DMRs from the first two steps. We will specifically determine if each of these dose-sensitive and/or cancer-related DMRs in former instructors more closely match the DNAm patterns of post-exposure than pre-exposure to test the hypothesis of stability of these biomarkers. We will further prioritize the most stable DMR biomarkers by this test.

Prioritizing best predictors of VOC and PAH exposure levels: Averaged DNAm levels will be calculated for each DMR prioritized from the first 3 steps. These will be correlated (by Pearson) with urine measurements of each VOC and PAH chemical from the same instructors/recruits and timepoints. We will also examine correlations between different chemicals, because highly correlated chemicals can serve as experimental replication to reduce false positive findings in correlations with DNAm data, as this is more effective than FDR correction for highly correlated measurements.

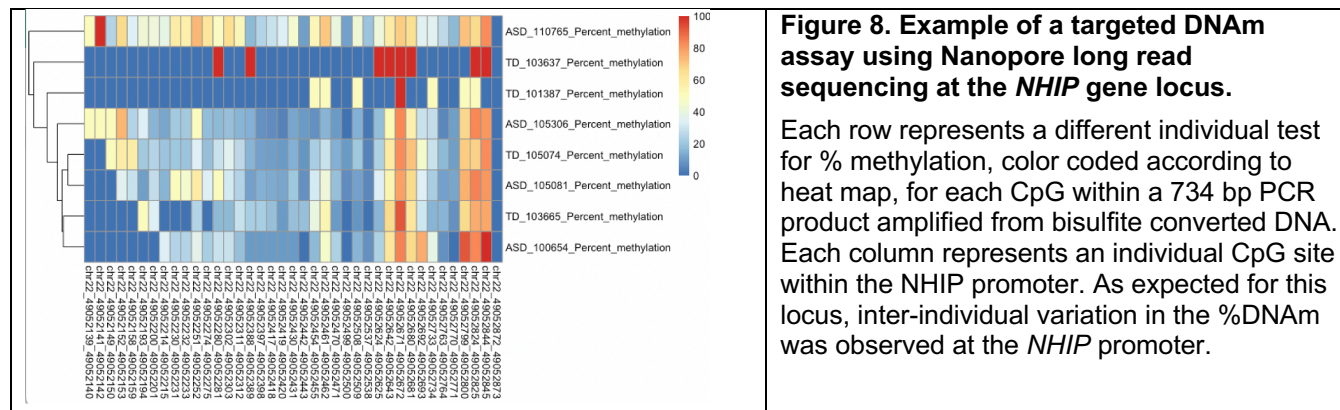
Replication with whole blood DNAm array data from Climate Action grant: By mid-late 2025, DNA samples isolated from whole blood for the 294 CAFF CRS subjects at two timepoints (baseline and 12 m) will have results from an Illumina EPIC array analysis of DNAm. The Climate Action grant proposes to test the hypothesis of epigenetic age acceleration. We propose to use this dataset to identify

potential overlap through region look-up analysis of our prioritized list of dosage-sensitive, cancer-relevant, stable, VOC and PAH-associated DMRs. We expect that some of our DMRs may replicate using a different DNA source and method, and these will be used for further prioritization of the ~10 DMRs to be tested in Aim C. Independently, we will also perform an multi-variant analysis on the CAFF CRS EPIC DNAm dataset using our recently developed tool Comethyl,⁵⁰ that is suitable to investigating more global DNA methylation patterns together with a large number of associated variables. In the CAFF CRS participants, these variables will include measurements from health assessments, clinical labs, body composition, risk factor questionnaires, food frequency questionnaires, patient-reported outcomes, physical activity and sleep monitoring, and exposure diaries.

Potential pitfalls and alternative approaches: The challenge with replication across WGBS and EPIC DNAm data is the EPIC array only covers ~2% of CpGs in the genome, so some regions will not be available for look-up. Therefore, we will not exclude any promising DMRs based on the lack of coverage in the EPIC dataset. If we identify additional covariates for the prioritized DMRs (such as BMI, sex, or other variable), we can adjust for them in the initial DMR analyses. There is no established method to determine statistical power for WGBS except to compare to prior studies, in which we have demonstrated statistically significant reproducible results using human sample sizes of <100^{34,51-54}, including the DMRs shown in Figure 4-5 which was only 51 human samples.

Aim 1C. FECS panel of ~10 prioritized DNAm loci will be tested for sensitivity in a new intervention cohort of instructors and recruits.

Development of a PCR-based targeted sequencing approach on a panel of prioritized DMRs: Because WGBS is likely too costly and data intensive for a routine health screening tool, we plan to test the efficacy of approximately 10 prioritized DMRs for targeted locus-specific sequencing. PCR primers will be designed to amplify bisulfite converted DNA over the prioritized DMRs, these PCR products will be gel purified and sent for bar-coded sequencing by either Nanopore (products >150 bp) or Illumina (<150 bp) platforms. We then quantify the converted (unmethylated) to unconverted cytosines in the PCR product and calculate % DNAm for each CpG and average DNAm levels over each DMR in each sample. An example of a successful quantitation of DNAm using locus-specific Nanopore sequencing is shown in Figure 8 for *NHIP* (neuronal hypoxia-induced, placenta associated), a novel ASD neuroprotective gene that Dr. LaSalle's lab discovered from WGBS data from placenta samples⁵³.



Perform FECS test on ~80 plasma samples collected pre- and post-exposure from instructor intervention in Aim 3: As part of Aim 3, the entire research team will work together to design possible interventions. One intervention, at the level of Administrative control, is predicted to be the limitation of instructors' training exercises to one exposure per day as opposed to the current 3-4 per day. Over the three-year period, and ~4 trainings per year, biospecimens will be collected pre- and post-training at the LACoFD. From these samples, we will select pre- and post- plasma samples from ~20 instructors who were limited to only 1 training per day for 5 days. Choosing from the remaining instructor biospecimens, we will choose an additional 20 participants from the typical exposure instructor group (3-4 trainings/day) as controls based on matching to the low exposure intervention by age, experience, sex, and BMI. We will isolate cfDNA from these samples and perform the FECS panel. Results will include quantitation of % methylation for each CpG and each gene locus of the 10 regions, as well as %

change in % methylation between pre- and post-exposure at each CpG and each gene. Each measurement will be compared between groups for significance by t-test, corrected for false discovery.

Machine learning prediction of FECS intervention results: The machine learning pipeline will include two algorithmically unique machine learning classifiers to inform final predictions, which include Random Forest and Support Vector Machine. The best performing model will then make final predictions. The area under the receiver operating characteristic curve (AUC) will be used as measure of how each model can effectively distinguish between different features of interest (pre- vs. post-, low vs. high) which will reflect the model performance and optimization across both sensitivity and specificity metrics.

Potential pitfalls and alternative approaches: We admit that aiming for a 10-locus panel is a somewhat arbitrary estimate, but it is based on cost both for this budget as well as what would be a cost-effective number of assays to perform that would be less expensive than WGBS. It is possible that fewer regions may have sufficient sensitivity to be both dosage-sensitive and predictive of cancer-relevant loci. If more than 10 regions are required, this would suggest that WGBS may be a better approach in the future, especially since sequencing costs are continuing to drop every year. There are additional machine learning algorithms (XGBoost, K-nearest-neighbors, neural network) that could be used if the prior two models are not successful.

Aim 2. Develop culturally competent protocols for report back of research findings to firefighters. (Mr. Sullivan and firefighters at LACoFD, in collaboration with UCD scientists)

Hypothesis: Culturally competent report back protocols for data from Aim 1 will be effective in impacting firefighters' perceived susceptibility and severity of occupational cancer risk and in overcoming perceived barriers to the implementation of preventative interventions to reduce cancer risk.

Rationale: Over the last decade, several fire service organizations such as fire departments, labor unions, and trade organizations have conducted educational campaigns in an attempt to raise awareness regarding firefighters' increased cancer risk and to convey best practices to firefighters for reducing their cancer risk. Despite these efforts, the development and implementation of data driven preventative interventions in the fire service has lagged behind⁵⁵⁻⁵⁷. This stems in large part from a knowledge gap among firefighters regarding the magnitude of their occupational exposures to carcinogenic agents and the associated biological effects. Consequently, the perceived cancer risk among firefighters does not match the abundance of scientific evidence available, and exposure reducing interventions, which are largely viewed by firefighters as being incompatible with their public safety duties, are not effectively implemented at the largest fire departments in California and across the United States⁵⁵⁻⁵⁸.

Health Belief Model (HBM): The HBM posits that an individual's health behaviors are driven by their perception of disease severity and their susceptibility to that disease, coupled with their perception of the barriers to and the benefits of adopting those behaviors^{59,60}. Hence, for firefighters to be motivated to develop and implement preventative interventions to reduce cancer risk (an occupational health behavior), they must accurately perceive the threat of occupational cancer associated with their exposure to products of combustion and the weight of this threat must outweigh their perception of barriers to developing and implementing preventative interventions.

Report Back Protocol: In order to accomplish the goal of impacting firefighters' perceived severity and susceptibility of occupational cancer risk associated with their exposures to products of combustion, we will develop a report back protocol for individual and aggregate urinary PAH and VOC findings and aggregate DNAm findings that achieves our communication goals and that is culturally competent in a way that meets the information needs of firefighter research participants⁶¹.

Quarterly research seminars: We propose to hold quarterly seminars over remote or hybrid platforms that will be led by both Co-PIs, Dr. Janine LaSalle and Mr. Thomas Sullivan, with assistance from Mr. Logan Williams (graduate student) and Ms. Emily Thrall (junior specialist), when appropriate. Ms. Thrall is the daughter of a Captain within the LACoFD and thus has a very helpful connection in her ability to translate our research findings to firefighters. At each seminar, Mr. Sullivan will present on the collection of biospecimens from the LACoFD and to which labs they have been sent for which research projects. Dr. LaSalle will present on the progress and stage of the research process (DNA isolation,

sequencing, bioinformatic analyses, PCR-based tests) as well as research results. Prior to each seminar, Dr. LaSalle will consult with Mr. Sullivan and Ms. Thrall about whether return of aggregate results on the FECS development would be considered culturally competent. We hope to be able to convey results about the cancer types that we are potentially identifying as future risks for firefighters' exposures from the DNA methylation patterns, the differences in these quantifiable measurements between recruits versus current or prior instructors, and the impact of the number of training exposures on the DNA methylation biomarkers that we develop. Some seminars could also include other UCD experts in Wildfire Health, including Dr. Keith Bein, who is building the simulated burn facility as part of the 2024 congressional earmark (described at the end of Section 2) because his research at this facility could be helpful in designing engineering-based interventions for firefighter trainings. At the end of each seminar, we will leave ample time to answer any questions from firefighters. We will provide digital and paper evaluation forms that ask questions about whether the information provided was useful and ask for suggestions about improving dissemination of findings as well as take ideas for data-driven interventions and suggestions for future research questions to address. Seminars will be recorded and posted online for firefighters who were not available to attend the seminars.

Information sheets (paper and digital): After each quarterly seminar, we will prepare a 1-page summary with graphics and bullet point summaries of the information.

Yearly short modular seminars timed with annual required EMS training: At the yearly EMS training of the LACoFD, we plan to piggyback these in person meeting with a short (5-7 min) seminar on the progress of research and any research findings presented in a culturally competent and digestible manner. Information sheets described above will be disseminated and a portal for asking questions, as well as providing suggestions and ideas for possible interventions will be provided.

This approach will include written material, graphics, figures and other media tailored to the fire service, accompanied by in person report back seminars for firefighters led by Firefighter Sullivan and Dr. LaSalle. In order to address challenges that may arise due to variability in numeracy, graph literacy, and health/biomedical research literacy among research participants⁶¹, we will develop our report back protocol collaboratively with firefighters from the California FIRESCOPE Cancer Prevention Subcommittee and the LACoFD's Training Services Section whose job duties include fire service cancer prevention.

Surveys: Pre-data report back surveys will be administered to research participants to assess components of the HBM as it relates to their perceived severity and susceptibility of occupational cancer risk and their perceived barriers to and benefits of implementing preventative interventions to reduce cancer risk. Post-data report back surveys will be administered immediately following in person report back seminars to measure the effectiveness of the report back protocol at impacting these components of the HBM. We will utilize McNemar's test to examine the mean difference in proportions between pre- and post-data report back.

Aim 3. Design and implement exposure reducing interventions that are compatible with firefighters' public safety duties and result in detectable differences in FECS biomarker panel. (entire team)

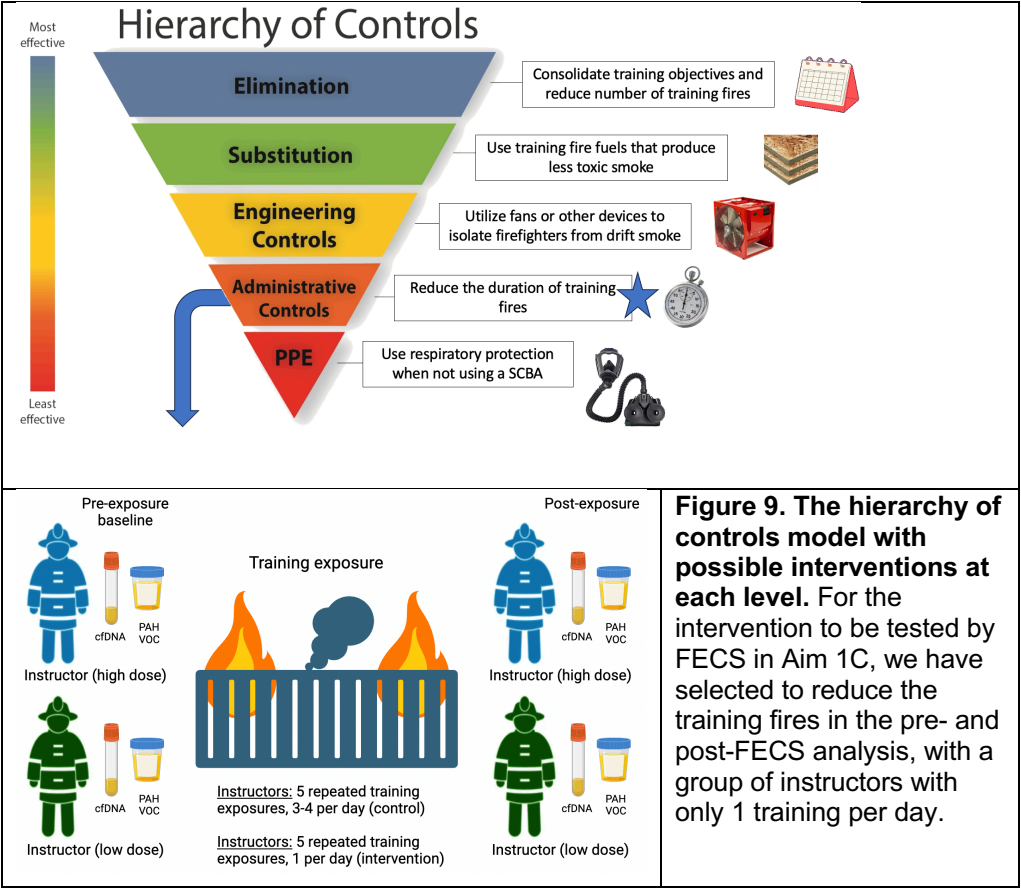
Hypothesis: Preventative interventions designed collaboratively by academic investigators and firefighters will be compatible with firefighters' public safety duties and effective as measured by implementation outcomes based on Enola Proctor's Model^{62,63}.

Rationale: Despite awareness of the increased cancer risk among firefighters, there exists the perception that preventative interventions are impractical, ineffective, and/or incompatible with public safety duties. As a result, best practices for exposure reduction are not effectively implemented at the largest fire departments in California and across the United States⁵⁵⁻⁵⁸.

Hierarchy of Controls: The NIOSH Hierarchy of Controls is an approach to exposure reduction designed to systematically identify measures that can be put in place to reduce firefighters' exposure to carcinogenic agents in both emergency and non-emergency settings. In order of decreasing effectiveness, the tiers of the Hierarchy of Controls include: Elimination, Substitution, Engineering Controls, Administrative Controls, and Personal Protective Equipment^{64,65}. **Figure 9** shows examples of

interventions at each level. We have chosen to use the Administrative control of reducing the duration of training fires to test as an intervention for the FECS biomarker panel being tested in Aim 1C.

Developing Preventative Interventions: In collaboration with firefighters from the FIREScope Cancer Prevention Subcommittee and the LACoFD’s Training Services Section, and utilizing data reported back in Aim 2, we will develop an array of exposure reducing interventions based on the Hierarchy of Controls. Interventions will be designed to be compatible with firefighters’ public safety duties and training objectives while also being effective at reducing exposures.



Implementation Outcomes: Utilizing validated instruments⁶⁶, we will engage LACoFD firefighters to evaluate the proposed preventative interventions by measuring the implementation outcomes of: acceptability (perception that interventions are agreeable, palatable, or satisfactory), feasibility (extent to which interventions can be successfully adopted), and appropriateness (perceived fit, relevance, and compatibility of interventions)^{56,57}. We will follow up with these firefighters after six months to evaluate the effectiveness of the proposed preventative interventions by measuring the implementation outcomes of: penetration (whether the interventions were successfully integrated), sustainability (whether interventions can be permanently adopted), and fidelity (whether interventions are implemented as designed)^{62,63}.

Dissemination: Research findings and interventions will be disseminated to the fire service at large through California’s FIREScope Cancer Prevention Subcommittee which was chartered to disseminate data driven best practices for reducing cancer risk in the fire service.

Table 1. Timeline	Year 1	Year 2	Year 3
Aim 1A WGBS/DMR 1B FECS panel 1C test FECS	Isolate DNA, WGBS	DMR generation FECS DMR selection FECS test development	FECS on intervention
Aim 2	Report back seminars on PAH and VOC	Report back seminars on DNAm findings	Report back seminars on intervention FECS
Aim 3		Develop instructor intervention	Collect samples pre post training

3. Partnership Collaboration Plan and Firefighter Benefit

Engagement of Community: The community of interest for this proposal is California firefighters because they face significant occupational risks for developing various types of cancer, due to prolonged exposure to carcinogenic substances encountered during urban and wildland firefighting activities, as well as simulated exposures during instruction and training. After discussions between UC Davis scientists and LACoFD firefighter researchers, we decided to focus on the exposures to products

of combustion that occurred during training of new recruits, because this was a controlled exposure dose and potentially the easiest to modify by interventions at multiple levels. Also, the interest of firefighters for the development of an early screening tool that was customized specifically for firefighters' exposures was discussed as a major unmet need of the community.

Plan for collaboration between LACoFD and UC Davis: The LACoFD Department Executive Staff has offered full support of this project. Firefighter Sullivan will be incorporating all department members involved in the study in pre and post sample seminars. The fire department FILM UNIT will be instrumental in the dissemination of information through department training websites as well as periodic informational videos regarding the study and any or all information as it is received. Dr. LaSalle and her research team will work directly with Firefighter Sullivan and the LACoFD to develop culturally competent data-driven information in the form of short videos, information sheets, and answering questions from firefighters about the research results. Dr. LaSalle plans to visit the LACoFD twice, and Emily Thrall (UCD researcher and daughter of a firefighter) will be working with Mr. Sullivan on site at the LACoFD approximately 4x per year. The entire research team will meet virtually at least once per month to discuss logistics of sample collections and shipment, research results, and any obstacles to progress. Outside of monthly group meetings, the research team will communicate by email and occasional phone calls. For this community-based research project the updated information will be instrumental to all participants and the LACoFD has shown they are willing to accommodate all needs. All sample collections will be performed at LACoFD Fire Stations and training facilities and the use of department administrative sites has been approved by the Fire Chief and his Executive Staff and Firefighter Sullivan will be assisted in specimen labelling, storage and shipments by Emily Thrall on her visits to LACoFD. At the UC Davis site, Dr. LaSalle will lead the research activities, including mentoring Mr. Williams and Ms. Thrall on the laboratory procedures and data analyses of Aim 1. Dr. Hussain will share DNAm and metadata from the CAFF CRS cohort and provide expertise in epidemiology and community-engaged research to the research project. As discussed in the Collaborative Agreements Form, all researchers on this team have agreed to collectively own the data and intellectual property rights from this project and we have a plan for handling disagreements if they arise.

Capacity building: We expect that the proposed research project will benefit everyone on the research team by capacity-building for skills such as performing team science and improving collaboration skills. The research trainees on the project, Mr. Williams and Ms. Thrall, will expand their knowledge of laboratory science from undergraduate and undergraduate education to include skills on how to apply science to solve specific problems in their community. Dr. LaSalle will improve her communication skills and learn about firefighting as a profession and the specific health concerns of firefighters to prompt new research ideas. Firefighter Sullivan will learn about the science of blood testing and how best to explain research results to the firefighting community. Dr. Hussain will learn about molecular biomarker testing design and implementation. Lastly, the entire research team is motivated to use data to prompt policy changes to improve cancer prevention in California firefighters.

5. Statement of Future Goals

At the end of this research collaboration, we expect to have obtained evidence that cfDNA DNAm analyses will reveal dosage-sensitive and cancer-relevant quantitative data that can be used to develop future cancer screening protocol for health monitoring of firefighters. By working together with firefighters in LACoFD, we will understand how to develop culturally competent report back protocols and the opinions of firefighters on receiving this information. Lastly, we will be able to provide information to administrators and policy-makers on predictive biomarkers of cancer risk, as well as the effects of data-driven interventions on these quantitative biomarkers.

We will likely need to apply for future funding to be able to develop the FECS biomarker panel for clinical utility in health screening of firefighters. In the future, we would be interested in expanding screening beyond cancer prevention, including reproductive impacts and risks to the children of firefighters in developing neurodevelopmental disorders.

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